

Mohamed Muaath Rifath

Software Engineer

me@muaathrifath.me • [+91 88837 35079](tel:+918883735079) • Chennai, India
muaathrifath.me • linkedin.com/in/muaath-rifath • github.com/muaath-rifath

PROFESSIONAL SUMMARY

Software Engineer focused on shipping zero-to-one products. I build systems across distributed Go backends, AI-integrated frontends, and custom ESP32 firmware. I prioritize reliability in core infrastructure like identity and payments, building with security and scalability in mind to support product growth. My focus is on solving the technical challenges of early-stage development to ensure a stable foundation for the product.

EXPERIENCE

Introspect Labs

Full Stack Developer

Oct 2025 – Feb 2026
Remote, Bengaluru, India

- Architected the v1 backend for uyir.ai, migrating the identity layer from legacy OTP to Zitadel OIDC; implemented automatic JWT refresh and patched critical Next.js CVEs mid-deployment.
- Engineered a **reconnection-aware attendance tracking system** for LiveKit video sessions using **event-sourced audit logs** and idempotency guards to handle webhook race conditions and network drops.
- Built the full Next.js 15 frontend from scratch with custom Tiptap diff marks for inline AI editing, therapist dashboards, and a **presigned S3 upload workflow** for AI avatar training data.
- Containerized the stack with Docker, set up self-hosted GitHub Actions CI/CD with Nginx reverse proxy, and migrated from SMTP to AWS SES for scalable transactional emails.
- Optimized database performance via PostgreSQL schema design and Alembic migrations, supporting complex therapist-user scheduling and multi-tier subscription billing.

PROJECTS

Sol, Open Distributed Smart Home & Agentic Orchestration Platform · Go 1.25, ESP32-S3, MQTT, LiveKit, MCP, TimescaleDB, MinIO, ZITADEL, mTLS

- Architected a 4-tier agentic platform featuring a Go control plane and ESP32-S3 nodes, validated through hardware deployment with per-device X.509 mTLS security.
- Engineered a **1.2s latency voice pipeline** using on-device wake-word detection (WakeNet9), LiveKit WebRTC streaming, and Azure OpenAI Realtime API for agentic reasoning.
- Hardened AI-to-hardware reliability via a **pgvector/HNSW grounding layer** that resolves natural language device references to UUIDs with sub-millisecond latency.
- Developed a zero-install **WebSerial provisioning flow** for browser-based firmware flashing and OIDC device-code registration, eliminating the need for local IDEs.
- Built a native MCP server with capability-based authorization, exposing a standard interface for **external agents (OpenClaw, NemoClaw, Claude)** to discover and actuate hardware.

Zylert, LLM-Orchestrated Notification Triage System · Kotlin, Android SDK 36, Room, Node.js, OpenAI/Mistral, Zod

- Architected an Android-to-Node.js pipeline that intercepts and triages push notifications in real time using **LLM-backed semantic classification**.
- Hardened LLM output reliability using **strict Zod validation** and fallbacks to handle malformed payloads and ensure 99% parsing accuracy.
- Engineered a priority routing system that surfaces critical human messages while compressing low-value automated noise via conversational querying.

Threble, Open-Source Social Network · Next.js 15, TypeScript, NextAuth, Prisma, PostgreSQL, Redux/RTK, Azure Blob

- Developed an open-source social platform featuring a LinkedIn-style graph with connection requests, mutual-connection lookups, and community roles.
- Implemented route protection with NextAuth, Prisma/PostgreSQL persistence, and responsive UI with mobile-aware notification flows.

SKILLS

Languages: TypeScript, Go, Python, Kotlin, C/C++ (ESP-IDF/Arduino), SQL, Shell/Bash, HTML/CSS

Frontend/Mobile: Next.js 16, React 19, D3.js, Three.js, R3F, Tailwind CSS, Zustand, Redux, Framer Motion, Android SDK

Backend/DevOps: FastAPI, Node.js, PostgreSQL (TimescaleDB/pgvector), MongoDB, Prisma, Celery, Redis, MQTT, Docker, Traefik, MinIO, GitHub Actions, AWS

AI/Security: OpenAI Realtime, LiveKit, RAG Pipelines, Zitadel OIDC, mTLS, X.509, OAuth 2.0, JWT, HMAC-SHA256

CERTIFICATIONS

- **MCP: Build Rich-Context AI Apps with Anthropic** – DeepLearning.AI (May 2025)
- **Docker Foundations Professional Certificate** – Docker, Inc. (Feb 2026)
- **Foundation of Cloud IoT Edge ML** – NPTEL (May 2024)

EDUCATION

Aalim Muhammed Salegh College of Engineering

B.E. in Electronics & Communication Engineering

2022 - 2026
Chennai, India